

Telephone network

Cell phones connect to base stations, each providing coverage of a hexagonal area called a cell. Each cell has a number of frequencies or channels available to callers. As cell phones each connect to a particular base station, the same frequencies can be used for callers using base stations elsewhere. Landline calls go through local and main exchanges.

Calling from a moving cell phone

User A's call is given a channel and routed via a base station to the mobile exchange. User A's phone checks the signal strength from nearby base stations, feeding this information back to the mobile exchange. It indicates the current signal is weakening as the caller leaves the cell.

Call handed over to new cell

The mobile exchange readsies a new channel for user A in the cell they are moving to and sends this information to user A's phone. User A's phone signals to the new base station its arrival in the new cell and the old channel is shut down.

Moving cellular call received

The mobile exchange scans for user B and puts through the call. B should not notice when A's signal is handed over.

Satellite phone

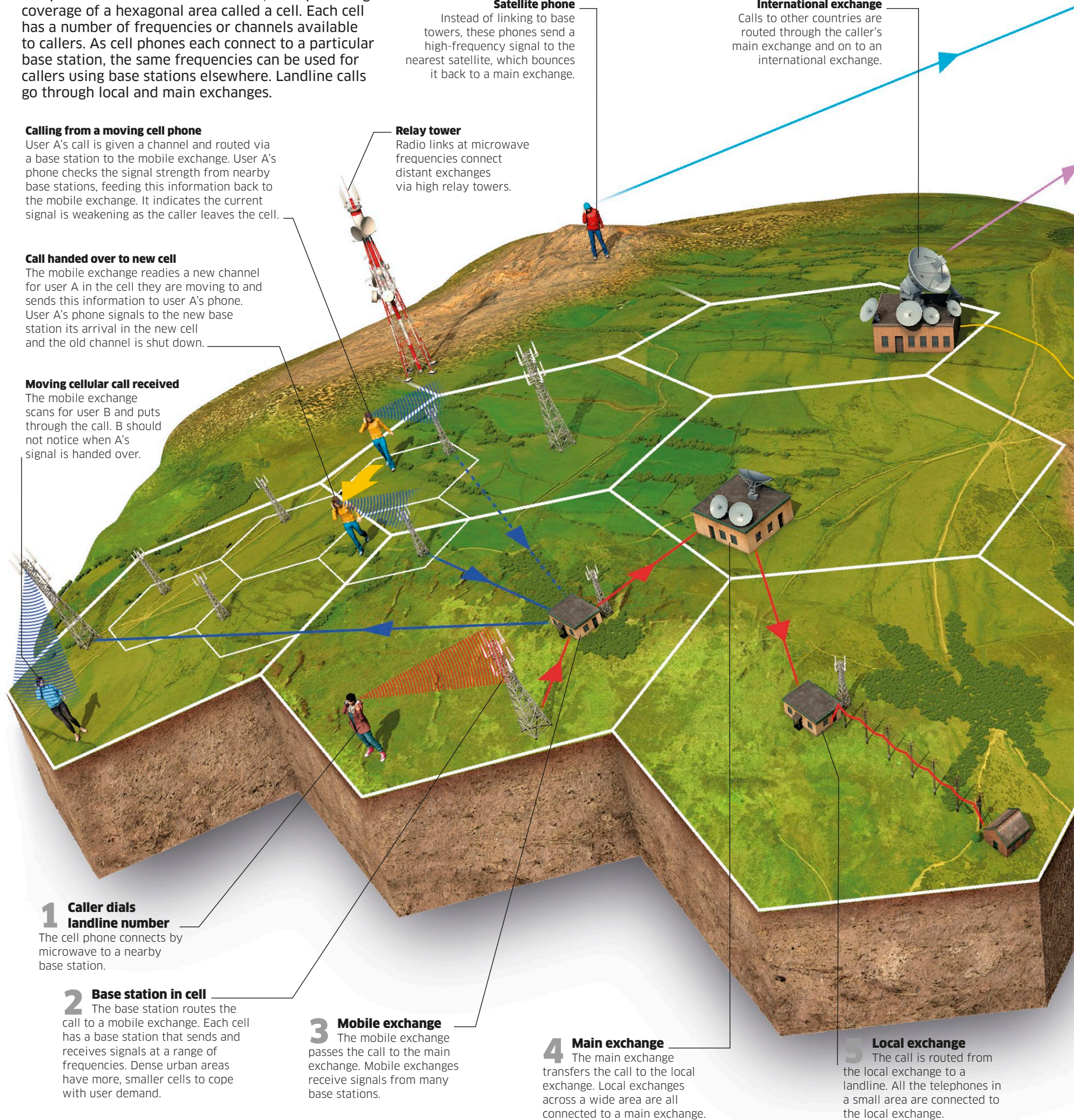
Instead of linking to base towers, these phones send a high-frequency signal to the nearest satellite, which bounces it back to a main exchange.

International exchange

Calls to other countries are routed through the caller's main exchange and on to an international exchange.

Relay tower

Radio links at microwave frequencies connect distant exchanges via high relay towers.



1 Caller dials landline number

The cell phone connects by microwave to a nearby base station.

2 Base station in cell

The base station routes the call to a mobile exchange. Each cell has a base station that sends and receives signals at a range of frequencies. Dense urban areas have more, smaller cells to cope with user demand.

3 Mobile exchange

The mobile exchange passes the call to the main exchange. Mobile exchanges receive signals from many base stations.

4 Main exchange

The main exchange transfers the call to the local exchange. Local exchanges across a wide area are all connected to a main exchange.

5 Local exchange

The call is routed from the local exchange to a landline. All the telephones in a small area are connected to the local exchange.